MADMEN

VIRAL(REAL) GROWTH MACHINE

What happened when we Reverse-engineered from 4,346 real data points

WHAT WE DID

4,346

Data points

3,353 Reddit + 993 YouTube

16

Analysis charts

3 tiers: descriptive, sentiment, virality

4-step

Processing pipeline

Clean → Features → NLP → LLM

5

Alert signals

Automated anomaly detection

HOW WE DID IT

REDDIT free public api **YOUTUBE** free public api

STEP PIPELINE

clean → feature engineering → sentiment analysis → LLM classification with Claude Sonnet 4

Analyzed the enriched data to find growth patterns

The LLM classified each post into content category, growth type, and target audience

Then we built an automated system that reruns the whole thing on schedule and alerts via Slack when something unusual happens. 4-step enrichment pipeline:

Data cleaning, Features of virality flags, NLP, LLM classification

Reddit public JSON YouTube Data API v3 Claude Sonnet 4 API VADER (nltk) TF-IDF (scikit-learn) matplotlib + seaborn Python + pandas

FINDING 1

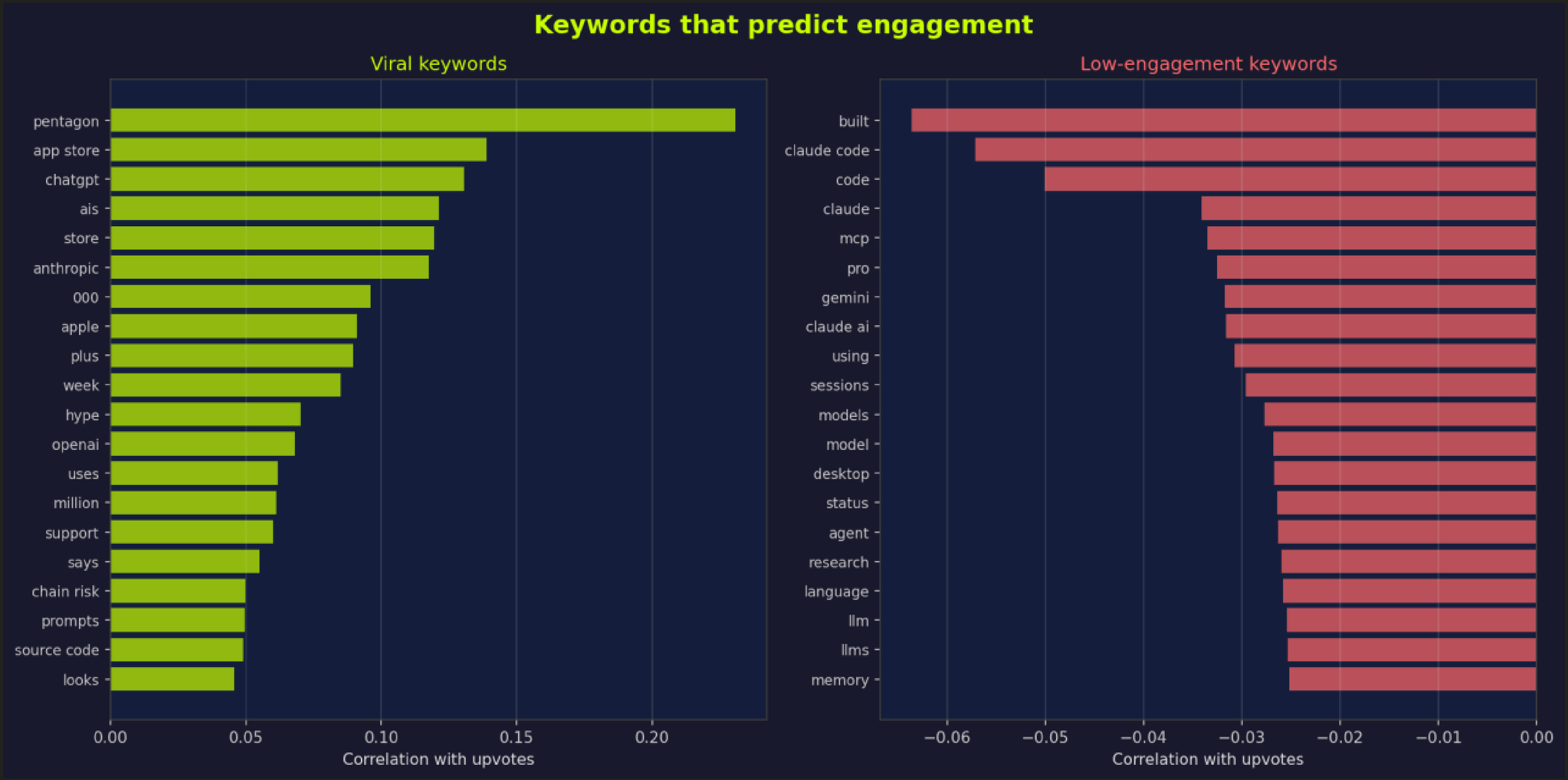
PENTAGON INCIDENT > EVERY PRODUCT LAUNCH

72
Pentagon posts

246K
Combined upvotes

3.8x
Above baseline

TOP POST: 39,102 UPVOTES ON R/TECHNOLOGY
CLAUDE HIT #1 ON APP STORE FROM A POLITICAL STANCE, NOT A FEATURE RELEASE.

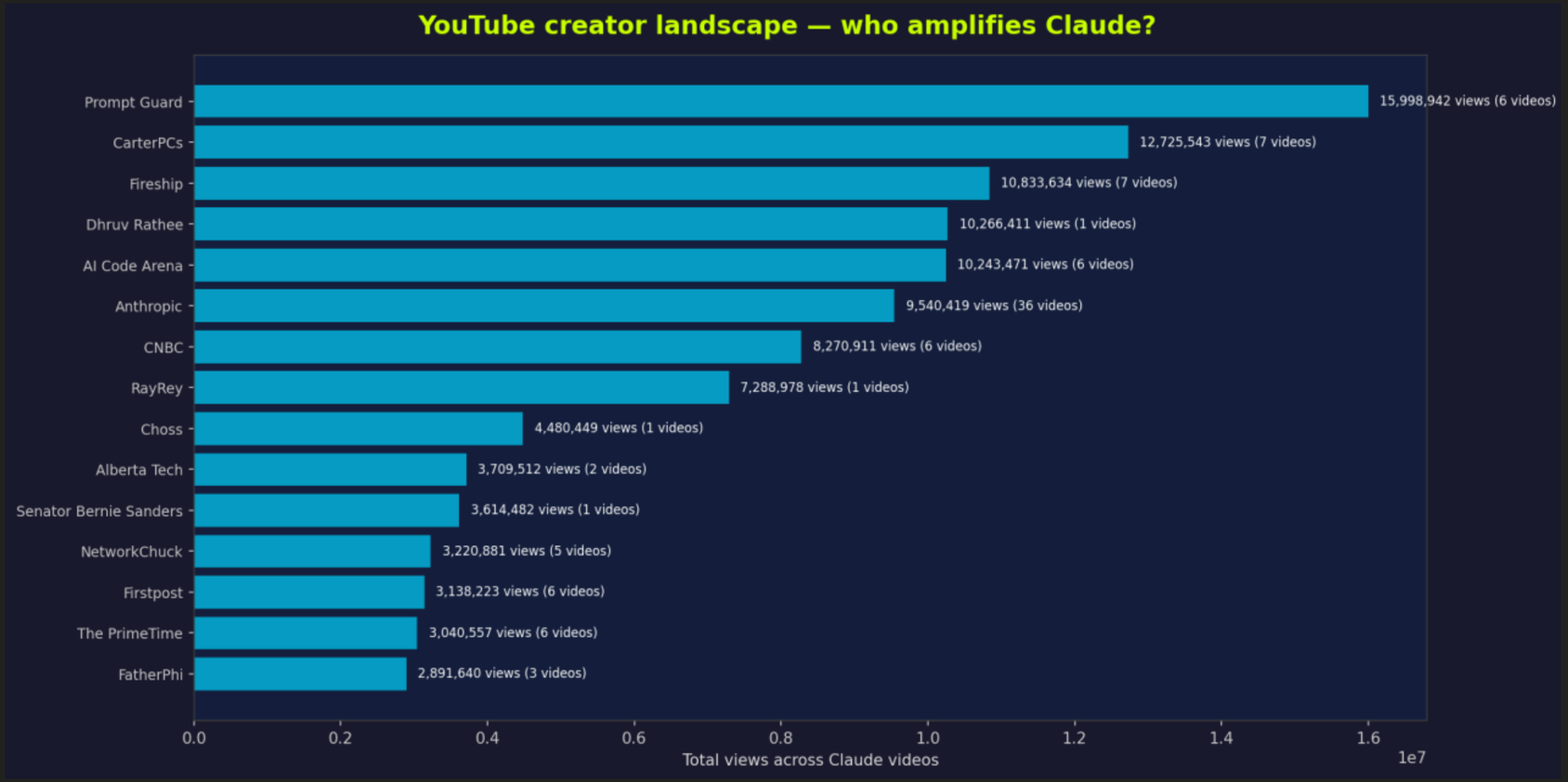

Keywords that predict engagement
Viral keywords
pentagon
app store
chatgpt
ais
store
anthropic
000
apple
plus
week
hype
openai
uses
million
support
says
chain risk
prompts
source code
looks
0.00
0.05
0.10
0.15
0.20
Correlation with upvotes
Low-engagement keywords
built
claude code
code
claude
mcp
pro
gemini
claude ai
using
sessions
models
model
desktop
status
agent
research
language
llm
llms
memory
−0.06
−0.05
−0.04
−0.03
−0.02
−0.01
0.00
Correlation with upvotes

FINDING #2

REACTIVE CONTENT BEATS EDUCATIONAL BY 2.7X

Reactive	1,681	Response to events
Competitive	1,675	vs competitors
Organic	997	User experiences
Educational	612	Tutorials, guides

3 CREATORS (PROMPT GUARD, CARTERPCS, FIRESHIP)
=
ANTHROPIC'S ENTIRE CHANNEL



FINDING #7

OTHER FINDINGS

Comments predict virality	0.67 correlation. Title formatting: ~0 correlation.
Complaints > praise	Negative content (1,462 avg) beats positive (1,342) for engagement.
Sunday = sweet spot	80% higher engagement than Thursday. Most content posted mid-week.
Viral posts flip audience	Daily community = developers. Viral moments = general public.

4,346 DATA POINTS
8 FINDINGS
6 STRATEGIES
1 AUTOMATED MACHINE

Counter-playbook

6 strategies for a competing product

01 Engineer a values moment
Pentagon = 246K upvotes

02 Seed mainstream communities
r/technology = 3.9x r/ClaudeAI

03 Two-tier creator partnerships
3 creators = Anthropic's channel

04 Build reactive content systems
Reactive = 2.7x educational

05 Optimize launch timing
Sunday = +80% engagement

06 Dual-engine growth system
Retention + acquisition

BUDGET ALLOCATION

CATEGORY	%	WHY
Creator partnerships	30%	3-8x more efficient than owned content
Reactive content team	25%	2.7x more engagement than tutorials
Community management	20%	Retention engine — amplifies viral moments
Event/narrative prep	15%	Pre-build assets for values moments
Analytics + monitoring	10%	Automated pipeline feeds everything above

218 batches, 4,346 posts, 544K tokens
Input: 299K tokens × $3/1M = $0.90
Output: 245K tokens × $15/1M = $3.68
Total: $4.58 per run
Monthly (4 runs): $18.32
At 10x scale (43K posts): $45.80/month

Every strategy traces to a specific data finding. Includes KPIs, budget allocation, and 6-week execution calendar.

218 API batches | ~530K tokens | All consumption tracked in llm_consumption.csv

Token Estimates for 3,357 Reddit posts:
- Input: ~305,000 tokens
- Output: ~121,000 tokens
- Total: ~425,000 tokens

COST ESTIMATES:
- GEMINI FLASH: $0.08
- GPT-4O MINI: $0.12
- CLAUDE SONNET: $2.72

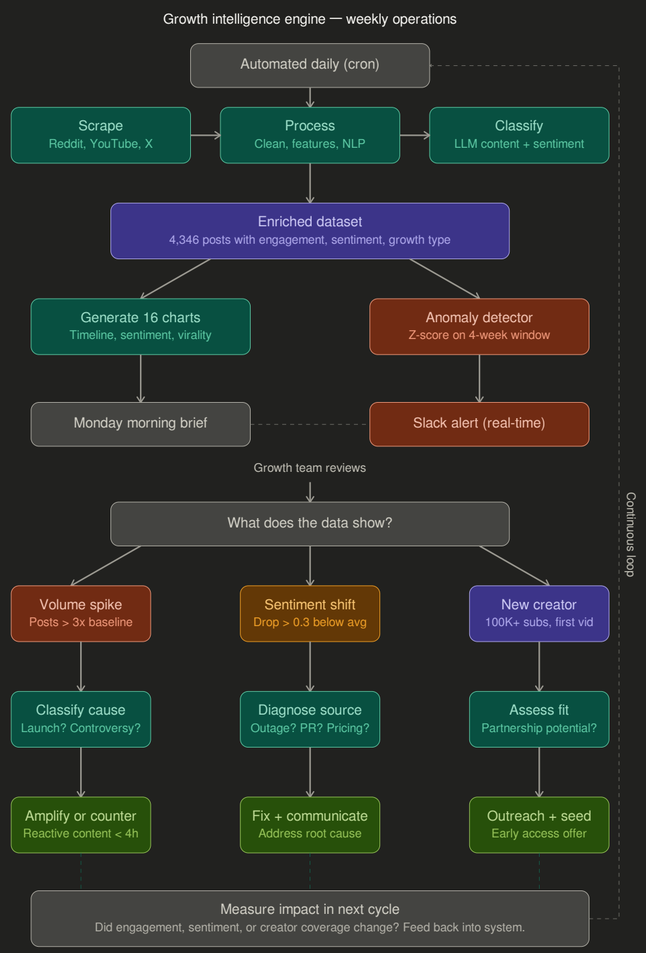
THE MACHINE

AUTOMATED GROWTH ENGINEER (+ORGANIC) SYSTEM

COST AT CURRENT SCALE: $0-5/MONTH (OLLAMA) | AT 10X: $10-70/MONTH

Fully working prototype: python machine.py --ai ollama --schedule 6h
Alerts: volume spike, sentiment crash, viral breakout, new creator, competitor surge
All code in repo, all data real, all pipeline runnable

Scrape	Process	Analyze	Alert	Act
Reddit + YouTube	4-step pipeline Clean → LLM	16 charts 3 analysis tiers	5 anomaly signals Slack webhook	Growth team Monday brief



hackathon-growth-app APP 8:12 AM

Growth Intelligence Alert — 2026-04-05 08:12

🟡 **MEDIUM**

- New creator entry: Felicia Simon (156,000 subscribers) posted first Claude video on 2026-03-30
  - Signal: new_creator
  - Platform: youtube
  - Dataset rows: 993
  - Threshold: 100000
  - Channel: Felicia Simon
  - Subscribers: 156000
  - First date: 2026-03-30T00:00:00
  - Action: Review in next standup and add to weekly growth brief.

⚪ **LOW**

- reddit competitor surge: 16 mentions this week vs 6 last week (2.7x, threshold=2.0x)
  - Signal: competitor_surge
  - Platform: reddit
  - Dataset rows: 3353
  - Threshold: 2.0
  - Ratio: 2.7
  - Current week mentions: 16
  - Previous week mentions: 6
  - Week ending: 2026-04-05T00:00:00
  - Action: Track trend; no immediate escalation required.

REFERENCES

HUTTO, C. J., & GILBERT, E. (2014). VADER: A PARSIMONIOUS RULE-BASED MODEL FOR SENTIMENT ANALYSIS OF SOCIAL MEDIA TEXT. PROCEEDINGS OF THE 8TH INTERNATIONAL AAAI CONFERENCE ON WEBLOGS AND SOCIAL MEDIA (ICWSM-14).

KIM, J., & PARK, E. (2025). GOING VIRAL: HOW SOCIAL AND PERSONAL MOTIVATIONS DRIVE EMOTIONAL ENGAGEMENT AND CONSUMER ONLINE BRAND-RELATED ACTIVITIES. PLOS ONE. HTTPS://DOI.ORG/10.1371/JOURNAL.PONE.0336907

SCHMIDT, E., DI MARCO, N., GIACOMO, E., CRESCI, S., CINELLI, M., & QUATTROCIOCCHI, W. (2025). EVALUATING THE EFFECT OF VIRAL POSTS ON SOCIAL MEDIA ENGAGEMENT. SCIENTIFIC REPORTS, 15. HTTPS://DOI.ORG/10.1038/S41598-024-84960-6

TOOLS & APIS

ANTHROPIC. (2026). CLAUDE API DOCUMENTATION. HTTPS://DOCS.ANTHROPIC.COM

GOOGLE. (2026). YOUTUBE DATA API V3. HTTPS://DEVELOPERS.GOOGLE.COM/YOUTUBE/V3

OLLAMA. (2026). OLLAMA: LOCAL LARGE LANGUAGE MODEL RUNTIME. HTTPS://OLLAMA.COM

PEDREGOSA, F., ET AL. (2011). SCIKIT-LEARN: MACHINE LEARNING IN PYTHON. JOURNAL OF MACHINE LEARNING RESEARCH, 12, 2825–2830.

REDDIT. (2026). REDDIT JSON API. HTTPS://WWW.REDDIT.COM/DEV/API